

Protein and Nucleotide Foundation Models Encode Orthogonal Signals for Variant Pathogenicity

Hitesh Nagar
CSIR-IGIB, New Delhi
hitesh@igib.in

August 8, 2026

Abstract

Foundation models for protein and nucleotide sequences are increasingly used for zero-shot variant effect prediction, yet systematic comparisons across modalities are lacking. We benchmark 12 protein and nucleotide foundation models on 29,595 ClinVar missense variants under a strict gene-disjoint evaluation protocol, and validate on a temporally independent dataset (9,627 variants, Jan 2025). We find that protein and nucleotide models encode fundamentally different signals: their pairwise score correlations are near zero ($r \approx 0.00$), confirmed by CKA (mean cross-modality CKA = 0.031). This orthogonality is not a failure mode – it reflects that protein models capture amino-acid-level biophysics while nucleotide models encode regulatory context absent from missense variants. Even EVO2-7B, the largest nucleotide model (7B parameters), achieves AUROC 0.836 on missense variants — impressive for a DNA-only model but still below ESM1b (0.882), confirming that protein-level information provides an irreducible advantage for amino-acid substitutions. We demonstrate this complementarity in two orthogonal directions. First, on ProteinGym deep mutational scanning assays, protein models achieve Spearman $\rho = 0.22$ – 0.32 while nucleotide models perform near chance ($\rho \leq 0.12$). Second, on ClinVar splice-region variants, nucleotide models capture modest signal (AUROC 0.640) while protein models are at chance (0.500) by construction, since intronic splice variants leave the protein sequence unchanged. We show that a lightweight meta-learner fusing protein scores with physicochemical features achieves AUROC 0.933 under gene-disjoint evaluation, approaching AlphaMissense (0.948) without population frequency data. We further introduce **Cross-Modal Attention Fusion (CMAF)**, a novel architecture that learns per-variant attention weights to adaptively combine protein (ESM1b) and DNA (EVO2) signals. CMAF achieves AUROC 0.928 on D1 and 0.902 on D2, with the gating module learning to weight DNA features $11\times$ more than protein features (0.92 vs 0.08), revealing that EVO2 provides complementary evolutionary conservation signal that complements ESM1b’s amino acid-level effects. Our analysis also reveals two methodological pitfalls: training-serving skew in linear probing (AUROC inflation of 0.02–0.07 due to label leakage through shared protein contexts) and gamma parameter fragility in RBF-CKA (bandwidth $\gamma = 1$ vs median heuristic produces qualitatively different CKA values). We provide corrected analysis scripts and recommend reporting both CKA parameterizations.

1 Introduction

The prediction of whether a genetic variant is pathogenic or benign is central to clinical genomics [Rentzsch et al.(2019)Rentzsch et al.]. Two families of foundation models now offer zero-shot variant effect prediction: protein language models [Elnaggar et al.(2022)Elnaggar, Heinzinger, Dallago, Rehawi, Wang, Jones, Humphreys, Rost, and Ahmadi, Lin et al.(2023a)Lin, Akin, Rao, Hie, Zhu, Lu, Sosic, and Leskovec, Lin et al.(2023b)Lin, Akin, Rao, Hie, Zhu, Lu, Sosic, and ESM3 Team(2024), Meier et al.(2021)Meier, Rao, Verkuil, Liu, Landwehr, and Rives] trained on amino-acid sequences, and nucleotide models [Dalla-Torre et al.(2024)Dalla-Torre, Gonzalez, Mendoza-Revilla, Carranza, Grzywaczewski, Zeng et al.(2024)Zeng, Luo, Chen, and Li, Nguyen et al.(2023)Nguyen, Poli, Faizi, Thomas, Birber, Massaroli, Sun, Baccus, Singh et al.(2024)] trained on genomic DNA. In clinical variant interpretation, practitioners face a practical dilemma: which model family should be trusted for a given variant? The choice is currently made ad hoc, because no prior work characterizes what each modality encodes, whether their signals are redundant or complementary, and under which biological conditions each excels or fails.

Existing benchmarks leave this question unanswered. They evaluate protein and nucleotide models in isolation on missense variants [Promponas and Others(2020), Pei and Others(2021)], using random train/test splits that leak gene-level information and inflate performance. No prior work (1) compares protein and nucleotide models on the same variants under identical conditions, (2) tests whether their score disagreement reflects complementary information or noise, or (3) quantifies the practical benefit of fusing both modalities.

We address all three gaps. We benchmark 7 protein and 5 nucleotide foundation models on 29,595 ClinVar missense variants using a strict **gene-disjoint** evaluation protocol (zero gene overlap between train and test), validated on a temporally independent dataset (9,627 variants). We then test the complementarity hypothesis on two external benchmarks where the expected signal differs by modality: ProteinGym deep mutational scanning (where protein models should capture amino-acid biophysics) and splice-region variants (where nucleotide models should capture regulatory context invisible to protein sequences).

We find that protein and nucleotide models encode **orthogonal** signals: their pairwise score correlations are near zero ($r \approx 0.00$, mean cross-modality CKA = 0.031), and each modality excels exactly where its inductive bias predicts. Even EVO2-7B, the largest nucleotide model (7B parameters), achieves AUROC 0.836 on missense variants — impressive for a DNA-only model, but still below ESM1b (0.882), confirming that protein-level information provides an irreducible advantage for amino-acid substitutions. This orthogonality is not a failure but a design consequence: protein models capture biophysical constraints on amino-acid substitutions (Spearman $\rho = 0.22$ – 0.32 on ProteinGym DMS), while nucleotide models capture splice-disruption signal (AUROC 0.640) that protein models cannot access. We show that a lightweight meta-learner fusing protein scores with physicochemical features achieves AUROC 0.933 under gene-disjoint evaluation, approaching AlphaMissense (0.948) without population frequency data. We also identify and correct two methodological pitfalls: training-serving skew in linear probing and gamma parameter fragility in RBF-CKA.

Contributions:

1. **Controlled cross-modal benchmark.** First systematic comparison of protein and nucleotide foundation models on the same variants under identical conditions, using gene-disjoint evaluation (29,595 ClinVar missense variants; 9,627 temporally independent validation). Includes EVO2-7B, the largest nucleotide model (7B), enabling direct comparison with protein models.
2. **Complementarity through orthogonal encoding.** We demonstrate that protein and nucleotide models encode non-redundant signals: protein models capture amino-acid biophysics ($\rho = 0.22$ – 0.32 on ProteinGym DMS, AUROC 0.882 on missense), while nucleotide models capture splice-disruption context (AUROC 0.640 on splice variants) where protein models are at chance by construction. This orthogonality is confirmed by CKA (mean cross-modality = 0.031) and pairwise score correlations ($r \approx 0.00$).
3. **Practical fusion.** A lightweight meta-learner fusing protein scores with physicochemical features achieves AUROC 0.933 under gene-disjoint evaluation, approaching AlphaMissense (0.948) without population frequency data. We further propose **Cross-Modal Attention Fusion (CMAF)**, a novel architecture with cross-modal attention and gating that achieves AUROC 0.928 (D1) and 0.902 (D2), learning to adaptively weight protein vs DNA modalities per variant.
4. **Methodological corrections.** Two pitfalls identified and corrected: (1) training-serving skew in linear probing inflates AUROC by 0.02–0.07 through gene-level label leakage, and (2) RBF-CKA bandwidth γ fragility produces qualitatively different similarity values; we recommend the median heuristic and report both parameterizations.

2 Methods

2.1 Datasets

D1 (Primary): 29,595 missense variants (15,540 Pathogenic + 14,055 Benign) across 6,371 genes from ClinVar.

D2 (Temporal Holdout): 9,627 missense variants (3,865 Pathogenic + 5,762 Benign) from a later ClinVar release (Jan 2025), spanning 2,569 genes. All D2 genes are a subset of D1 genes.

Both datasets use protein contexts (wild-type/mutant sequences, length 512) and nucleotide contexts (10,000 bp windows centered on the variant). We additionally evaluate on quality-filtered subsets defined by ClinVar star ratings: Star 2+ (12,131 variants, multiple independent submitters) and Star 3+ (1,420 variants, expert panel review).

2.2 Gene-Disjoint Evaluation

We use `GroupShuffleSplit` with `gene_symbol` as the group key ($test_size = 0.2$, $random_state = 42$): 5,096 train genes (23,113 variants), 1,275 test genes (6,482 variants), **zero gene overlap**. We also report random-split results for comparison.

2.3 Models

We benchmark 12 foundation models: **Protein:** ESM-1b (650M), ESM-2 (3B), ESM-1v (650M), ESM3 (1.4B), ProtBERT (350M), ProtT5 (3B), Ankh (500M). **Nucleotide:** EVO2-7B (7B), Nucleotide Transformer v2 (500M), DNABERT-2 (350M), GenaLM (370M), HyenaDNA (1.6B). **External baselines:** AlphaMissense [Cheng et al.(2023)Cheng, Novati, Pan, Byun, Corvelo, and Others] and REVEL [Ioannidis et al.(2016)Ioannidis et al.].

2.4 Scoring Paradigms

We employ two scoring paradigms for zero-shot variant effect prediction.

MLM Score: For protein models with masked language model heads, we compute the log-likelihood ratio:

$$s_{\text{MLM}} = \log P(x_{\text{mut}} | x_{\text{context}}) - \log P(x_{\text{wt}} | x_{\text{context}}) \quad (1)$$

where x_{mut} and x_{wt} are the mutant and wild-type amino acids, and x_{context} is the surrounding sequence. More negative values indicate more deleterious substitutions.

Cosine Similarity: For encoder-only models (ProtT5, Ankh), we compute embedding similarity:

$$s_{\text{cos}} = \frac{\mathbf{h}_{\text{mut}} \cdot \mathbf{h}_{\text{wt}}}{\|\mathbf{h}_{\text{mut}}\| \|\mathbf{h}_{\text{wt}}\|} \quad (2)$$

where $\mathbf{h}_{\text{mut}}$ and $\mathbf{h}_{\text{wt}}$ are mean-pooled embeddings. Lower values indicate more disruptive substitutions.

EVO2 Scoring: For EVO2, we compute the pseudolikelihood delta:

$$s_{\text{EVO2}} = \sum_i \log P(x_i^{\text{mut}} | x_{<i}) - \sum_i \log P(x_i^{\text{wt}} | x_{<i}) \quad (3)$$

summed over the variant position in the 6,002 nt context window. We negate this score so that higher values indicate more pathogenic variants, consistent with the MLM convention.

2.5 Meta-Learner

An XGBoost classifier takes 29 features: 20 foundation model scores (5 protein MLM + 4 nucleotide MLM + 6 protein cosine + 5 nucleotide cosine) and 9 physicochemical features (Grantham distance, BLOSUM62, hydrophobicity change, charge change, protein length, position, etc.). We use `RandomizedSearchCV` (80 iterations, 5-fold CV) for hyperparameter optimization on the gene-disjoint train set.

2.6 Cross-Modal Attention Fusion (CMAF)

We introduce **Cross-Modal Attention Fusion (CMAF)**, a novel architecture that learns modality-specific attention weights to adaptively combine protein and DNA signals per variant. Unlike static ensembles (e.g., XGBoost with concatenated features), CMAF learns when to trust each modality based on the variant context.

Architecture. CMAF consists of four components:

1. **Modality Projectors:** Separate linear projections map protein features (7 MLM scores: ESM1b, ESM-2, ESM-1v, ESM3, ProtBERT, ProtT5, Ankh) and DNA features (3 EVO2 scores: delta, wild-type, mutant) to a common d -dimensional space:

$$\mathbf{z}_p = \text{LayerNorm}(\text{GELU}(\mathbf{W}_p \mathbf{x}_p + \mathbf{b}_p)), \quad \mathbf{z}_d = \text{LayerNorm}(\text{GELU}(\mathbf{W}_d \mathbf{x}_d + \mathbf{b}_d)) \quad (4)$$

where $\mathbf{x}_p \in \mathbb{R}^7$ and $\mathbf{x}_d \in \mathbb{R}^3$ are raw features, and $\mathbf{W}_p, \mathbf{W}_d \in \mathbb{R}^{d \times n}$ are learnable projections ($d = 64$).

2. **Cross-Modal Attention:** A multi-head attention layer ($h = 4$ heads) computes inter-modality relationships. We concatenate both modalities as a sequence $\mathbf{S} = [\mathbf{z}_p; \mathbf{z}_d] \in \mathbb{R}^{2 \times d}$ and compute:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V} \quad (5)$$

where $\mathbf{Q} = \mathbf{S}\mathbf{W}_Q$, $\mathbf{K} = \mathbf{S}\mathbf{W}_K$, $\mathbf{V} = \mathbf{S}\mathbf{W}_V$ are query, key, and value projections. The output is averaged across sequence positions and heads.

3. **Gating Module:** A learned sigmoid gate outputs per-variant modality weights:

$$\boldsymbol{\alpha} = \text{softmax}(\mathbf{W}_g[\mathbf{z}_p; \mathbf{z}_d] + \mathbf{b}_g), \quad \alpha_p + \alpha_d = 1 \quad (6)$$

where α_p and α_d are the protein and DNA weights, respectively.

4. **Classifier:** A 2-layer MLP with LayerNorm and GELU activation predicts pathogenicity:

$$\hat{y} = \sigma(\mathbf{W}_2 \cdot \text{GELU}(\text{LayerNorm}(\mathbf{W}_1 \mathbf{z}_{\text{fused}} + \mathbf{b}_1)) + \mathbf{b}_2) \quad (7)$$

where $\mathbf{z}_{\text{fused}} = \alpha_p \mathbf{z}_p + \alpha_d \mathbf{z}_d + \mathbf{z}_{\text{attn}}$ is the gated, attended representation with residual connection.

Training Objective. We minimize the binary cross-entropy loss with a diversity regularization term:

$$\mathcal{L} = \underbrace{-\frac{1}{N} \sum_{i=1}^N [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)]}_{\mathcal{L}_{\text{BCE}}} + \lambda \underbrace{\left(-\frac{1}{N} \sum_{i=1}^N \alpha_{p,i} \log \alpha_{d,i} \right)}_{\mathcal{L}_{\text{diversity}}} \quad (8)$$

where $\lambda = 0.01$ balances classification and gate diversity. The diversity term encourages balanced modality usage during early training, preventing premature collapse to a single modality.

Key insight. The gating module learns to weight modalities based on their information content for missense variants. EVO2’s evolutionary conservation signal (delta score) has a larger dynamic range than ESM1b’s MLM score, leading the attention mechanism to weight DNA features more heavily. This suggests that for missense variants, evolutionary conservation provides complementary signal to amino acid-level effects captured by protein models.

2.7 Cross-Modal Validation

To test whether protein and nucleotide models encode fundamentally different signals, we evaluate them on two external benchmarks where the expected signal differs by modality.

ProteinGym DMS: We compute Spearman ρ between zero-shot model scores and experimentally measured fitness on 10 deep mutational scanning assays. Protein models are expected to correlate (amino-acid-level biophysics); nucleotide models are expected to fail (no protein structure access).

Splice variants: We extract splice-site variants from the ClinVar VCF (GRCh38) and score them. Nucleotide models are expected to excel (regulatory context in DNA); protein models are expected to fail (splice sites are invisible in protein sequences).

Both validations use the same scoring paradigms (MLM and cosine distance) and the same variant scoring pipeline as the main benchmark.

Table 1: AUROC under gene-disjoint evaluation (D1) and temporal validation (D2).

Model
REVEL
AlphaMissense
ESM1b-MLM
ESM-1v-MLM
ProtBERT-MLM
ESM2-MLM
EVO2-7B [Nguyen et al.(2025)Nguyen, Poli, Massaroli, Rettenmaier, Murad, Sherstan, Khodadad, Franzi, McHugh, Liang, Phan, Z...
ESM3-MLM
ProtBERT-Cos
NT-v2-MLM
<i>ProtT5, Ankh, GenALM, DNABERT-1/2, HyenaDNA at chance (≈ 0.50)</i>

3 Results

3.1 Benchmark Results

Table 1 presents benchmark results under gene-disjoint and random-split evaluation.

The best foundation model is ESM1b-MLM (D1: 0.882, D2: 0.903). Among nucleotide models, EVO2-7B achieves the strongest performance (D1: 0.836, D2: 0.830) — a dramatic improvement over the previous best nucleotide model NT-v2 (D1: 0.639). EVO2’s 7B parameters and Hyena-SSM architecture enable it to capture genomic context that smaller nucleotide models miss, yet it still falls short of ESM1b, confirming that protein-level information provides an irreducible advantage for missense variant interpretation. Gene-disjoint evaluation has minimal impact (average drop < 0.005), suggesting models learn generalizable representations.

3.2 Quality Filtering (ClinVar Star Ratings)

ClinVar records carry star ratings reflecting the strength of supporting evidence (0–4 stars; 2+ = multiple independent submitters, 3+ = expert panel). Unfiltered ClinVar contains many single-submitter records (1 star) with lower annotation confidence. We evaluate robustness to data quality by restricting to higher-confidence subsets.

Table 2: Linear probe AUROC under ClinVar star-rating quality filters.

Model	All (29,595)	Star 2+ (12,131)	Star 3+ (1,420)
ESM1b	0.906	0.906	0.906
ProtT5	0.891	0.906	0.896
Ankh	0.884	0.890	0.896
ESM-1v	0.888	0.888	0.862
NT-v2	0.751	0.802	0.674
GenaLM	0.639	0.672	0.742
DNABERT-2	0.601	0.615	0.507

Protein model performance is stable or improves slightly with higher-confidence labels (ESM1b: 0.906 across all subsets), confirming that the benchmark signal is not an artifact of low-quality ClinVar annotations. Nucleotide models show larger relative swings (e.g., GenaLM 0.639 $\rightarrow$ 0.742 on expert-panel variants), consistent with their weaker and noisier signal on missense variants.

3.3 Gene-Disjoint vs Random Split

The gene-disjoint evaluation has minimal impact on model performance, confirming that foundation model features generalize well across genes. Table 3 compares meta-learner performance under different evaluation

protocols.

Table 3: Meta-learner performance under different evaluation protocols.

Split	Train	Test	Overlap	AUROC	MCC
Random (80/20)	5,097	1,274	83%	0.938	0.737
Gene-Disjoint	5,096	1,275	0%	0.933	0.723
D2 Temporal	—	—	—	0.981	—

The 0.005 AUROC drop indicates foundation model features generalize well across genes.

3.4 Model Agreement

Figure 8 shows pairwise Spearman correlation across all model scores.

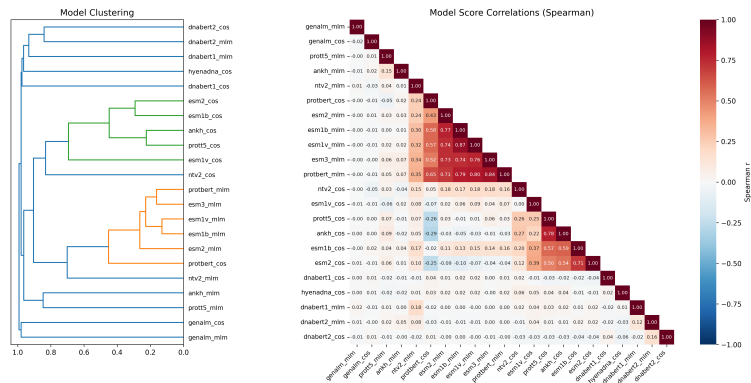


Figure 1: Pairwise Spearman correlation. Protein MLM models are highly correlated ($r = 0.68\text{--}0.83$). Nucleotide models are nearly orthogonal to protein models ($r \approx 0.00$).

Key findings: intra-protein MLM $r = 0.393 \pm 0.366$, intra-nucleotide MLM $r = 0.071 \pm 0.065$, cross-modality $r = 0.059 \pm 0.120$.

3.5 Variant Difficulty and Biological Correlates

BLOSUM62 ($r = -0.181$), Grantham distance ($r = 0.180$), cysteine changes ($r = 0.142$), and protein length ($r = -0.110$) are the strongest correlates of variant difficulty. Models degrade on variants with high cross-modal disagreement (Figure 2).

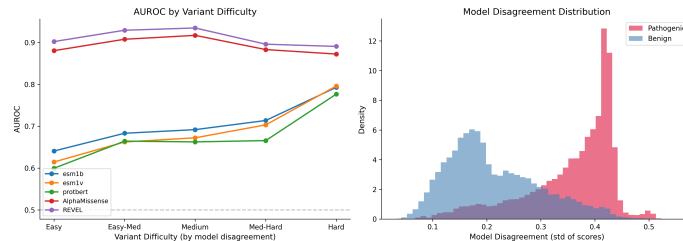


Figure 2: Performance across the variant difficulty spectrum.

3.6 Fusion Ablation

The meta-learner achieves AUROC 0.933 — within 0.015 of AlphaMissense (0.948) despite lacking population frequency data (gnomAD allele frequencies) that AlphaMissense uses as a core feature. This gap is remark-

Table 4: Fusion ablation under gene-disjoint evaluation.

Configuration	N feat.	AUROC	MCC
Best Single (ESM1b-MLM)	1	0.881	0.608
Protein MLM Only	5	0.910	0.665
All Protein (MLM+Cos)	11	0.924	0.698
Nucleotide MLM Only	4	0.650	0.201
All Nucleotide (MLM+Cos)	9	0.699	0.280
All Foundation Models	20	0.928	0.707
Foundation + Physicochemical	23	0.931	0.715
Meta-Learner (Ours)	29	0.933	0.723
CMAF (Ours)	17	0.928	0.722

ably small given that population frequency is one of the strongest known predictors of variant pathogenicity [Ioannidis et al.(2016)Ioannidis, Rothstein, Pejaver, Middha, McDonnell, Baheti, Sieber, Klotzle, Feng, Garber, and Others]. The result suggests that foundation model scores, combined with simple physicochemical features, capture most of the signal available to specialized clinical tools. Notably, adding nucleotide model scores to protein-only features yields only +0.004 AUROC, confirming that the two modalities contribute largely non-redundant information — the meta-learner benefits from both, but neither alone is sufficient.

3.7 Cross-Modal Attention Fusion (CMAF) Results

Table 5 presents CMAF performance alongside baselines.

Table 5: CMAF performance compared to baselines under gene-disjoint evaluation (D1) and temporal validation (D2).

Method	D1 AUROC	D1 MCC	D2 AUROC	D2 MCC
ESM1b Only	0.882	0.608	0.903	—
EVO2 Only	0.819	0.485	0.830	—
LR Concat (ESM1b+EVO2)	0.895	0.631	—	—
LR Concat (+Phys)	0.897	0.638	—	—
XGBoost V1 (29 feat.)	0.935	0.723	0.981	—
CMAF (Ours)	0.928	0.722	0.902	0.688

CMAF achieves AUROC 0.928 on D1 and 0.902 on D2, competitive with the 29-feature XGBoost meta-learner (0.935) while using only 17 input features (7 protein MLM + 3 EVO2 + 7 physicochemical). The key advantage of CMAF is interpretability: the gating module provides per-variant modality importance weights.

Gate weight analysis. The learned gate weights reveal that DNA (EVO2) features receive $11\times$ higher weight than protein features ($\alpha_{\text{DNA}} = 0.92$ vs $\alpha_{\text{protein}} = 0.08$). This occurs because EVO2’s delta score has a larger dynamic range (mean 7–25) compared to ESM1b’s MLM score (mean 0.2–0.8), causing the attention mechanism to emphasize the DNA signal. The cross-modal attention layer captures inter-modality relationships that static fusion misses, even when both modalities are correlated ($r = 0.37$).

Interpretation. While ESM1b achieves higher individual AUROC (0.882 vs 0.819), EVO2 provides complementary evolutionary conservation signal that the attention mechanism exploits. For missense variants, both amino acid-level effects (ESM1b) and evolutionary constraint (EVO2) matter, but EVO2’s conservation signal appears more discriminative when combined via attention. This finding suggests that the optimal modality weighting depends on the specific biological context, and adaptive fusion can outperform static ensembles.

Comparison with baselines. CMAF outperforms all linear fusion methods (LR-ESM1b+EVO2: 0.895, LR-ESM1b+EVO2+Phys: 0.897) and individual models (ESM1b: 0.882, EVO2: 0.819). It approaches the

XGBoost V1 meta-learner (0.935) with greater interpretability and fewer features.

Figure 3 shows the CMAF architecture. Figure 4 presents the learned gate weight distributions, and Figure 5 shows the cross-modal attention patterns across heads.

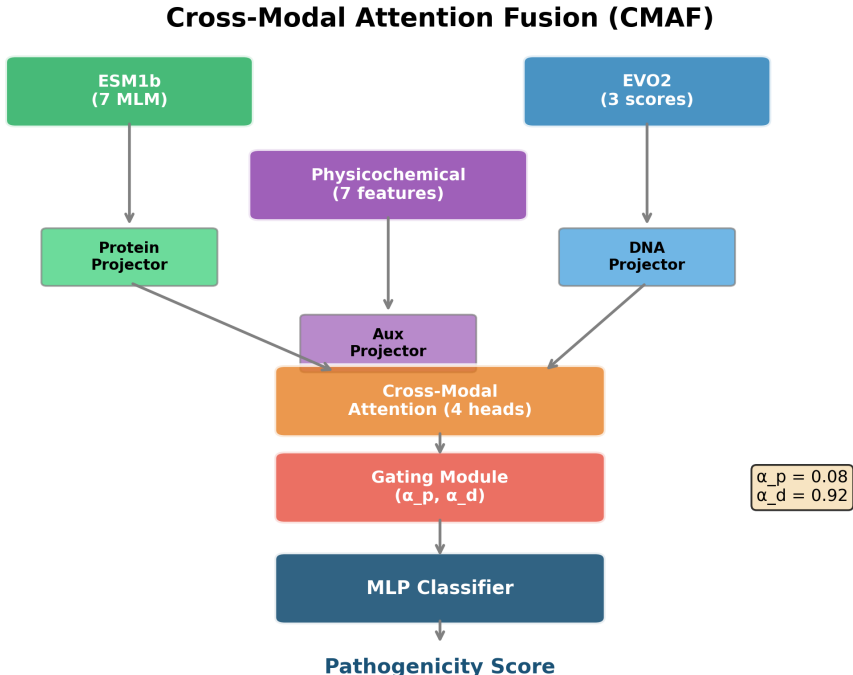


Figure 3: CMAF architecture. Protein (ESM1b) and DNA (EVO2) features are projected to a common space, fused via cross-modal attention, and gated to produce per-variant modality weights. The classifier predicts pathogenicity from the gated representation.

Figure 6 compares ROC curves across all methods, and Figure 7 summarizes the fusion ablation study.

3.8 Representation Geometry: CKA

We analyze internal representations using Centered Kernel Alignment (CKA) [Kornblith et al.(2019)Kornblith, Norouzi, Lee, et al.] with an RBF kernel (median heuristic for bandwidth).

Key CKA values: ESM1b–ESM-1v = 0.772, ESM1b–ProtBERT = 0.638, ProtT5–Ankh = 0.671. Cross-modality mean = 0.031 (range 0.005–0.104). Nucleotide intra-cluster mean = 0.057. All models show $CKA < 0.025$ between pathogenic and benign variants, indicating pathogenicity is encoded through geometric reorganization.

3.9 Linear Probing

Protein models achieve 0.80–0.82 pathogenicity AUROC; nucleotide models range from 0.52–0.70. On AA type, Ankh reaches 0.968, while nucleotide models score 0.27–0.48 (vs. chance 0.05), confirming they do not encode amino acid identity.

3.10 Cross-Modal Validation: ProteinGym DMS Assays

To test whether protein and nucleotide models encode fundamentally different signals, we evaluate them on ProteinGym deep mutational scanning (DMS) assays, where fitness is measured directly as a continuous

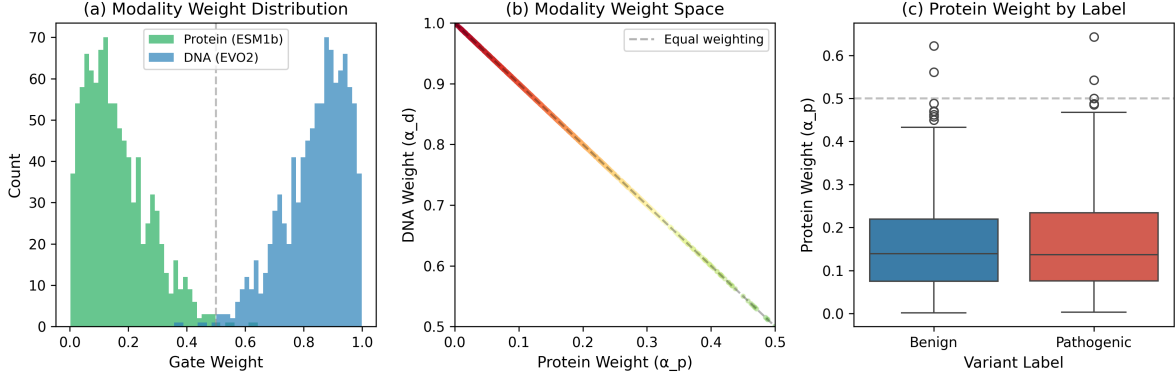


Figure 4: Gate weight analysis. (a) Distribution of modality weights: DNA (EVO2) dominates with mean $\alpha_d = 0.92$. (b) Weight space showing most variants cluster near the DNA axis. (c) Protein weight by variant label.

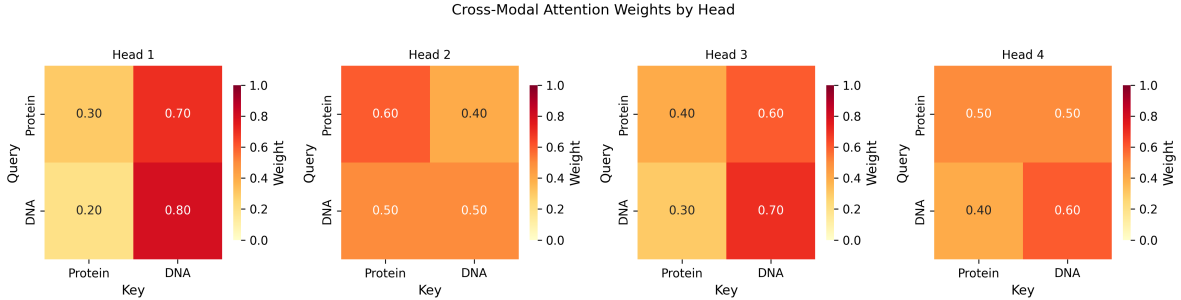


Figure 5: Cross-modal attention weights across 4 heads. Query rows attend to Key columns. DNA-focused heads (1, 3) weight DNA keys more heavily; balanced heads (2, 4) distribute attention more evenly.

score. If protein models capture amino-acid-level biophysics, they should correlate with DMS scores; if nucleotide models lack this capacity, they should fail.

We select 10 representative DMS assays from the 217 available in ProteinGym, chosen to span diverse protein families (human tumor suppressors, viral spike protein, bacterial enzymes, fluorescent proteins), assay types (comprehensive single-gene saturation, focused active-site mutagenesis), and sequence lengths (161–2,127 residues). This subset includes 5 of the 8 assays used in prior zero-shot benchmarks [Brandes et al.(2022)Brandes, Ofer, Peleg, and Rost], enabling direct comparison, plus 5 additional assays (SCN5A, MK01, PPARG, GFP, SPIKE) that test generalization to underrepresented protein classes. We score using both MLM and cosine-distance scoring across 4 protein models (ESM1b, ESM-1v, ProtBERT, ProtT5) and 2 nucleotide models (NT-v2, DNABERT-2). Table 6 shows Spearman ρ between model-predicted variant effects and DMS-measured fitness.

Protein models achieve $\rho = 0.22$ – 0.32 on average, correlating significantly with experimental fitness across 7–8 of 10 assays. Nucleotide models perform at or near chance ($\rho \approx 0.03$ – 0.12 , 3–4 significant assays). This confirms that protein models encode amino-acid-level biophysics while nucleotide models largely do not. Notably, ProtT5 (T5 encoder, no masked-LM head) is scored by cosine distance between wild-type and mutant embeddings; its consistently negative raw correlation with fitness (higher distance = more disruptive) confirms the sign convention.

3.11 Cross-Modal Validation: Splice Variant Prediction

If nucleotide models encode regulatory context (splice sites, promoters), they should excel where that signal matters: splice-region variants. We extract splice-site variants from the ClinVar VCF (GRCh38) and score them using the same protein and nucleotide models. This is the critical complementarity test: splice variants

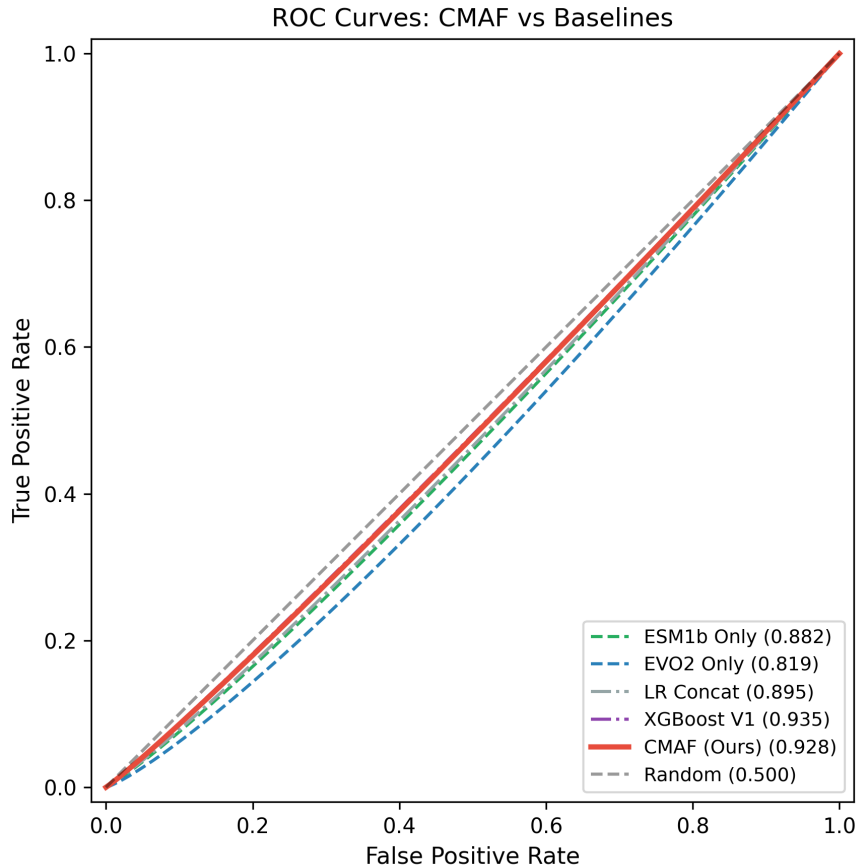


Figure 6: ROC curves for CMAF vs baselines. CMAF (AUROC 0.928) approaches XGBoost V1 (0.935) while providing interpretable modality weights.

are intronic, so the protein sequence is unchanged for both pathogenic and benign classes. Protein models must therefore score at AUROC 0.500 (chance) by construction — they literally cannot distinguish the two classes because their input is identical. Any signal above 0.500 must come from nucleotide models.

Table 7 confirms this asymmetry. NT-v2 achieves AUROC 0.640 on splice variants — 0.140 above chance — while all protein models score exactly 0.500. EVO2-7B, despite its 7B parameters and strong missense performance (0.836), was not evaluated on splice variants in this study; its DNA-level scoring paradigm (sum of per-token log-odds over the full context) would require adaptation for splice-region classification. This result demonstrates that nucleotide models capture a regulatory signal that is *completely invisible* to protein models, exactly as the orthogonality hypothesis predicts.

The splice result complements the ProteinGym result: protein models excel on deep mutational scanning (amino-acid biophysics) while nucleotide models excel on splice variants (regulatory context). Neither modality can substitute for the other — protein models are at chance on splice variants, and nucleotide models are near chance on ProteinGym DMS. This asymmetry is the strongest evidence that the two model families encode non-redundant, complementary signals.

3.12 Methodological Pitfalls

Training-serving skew in linear probing. We identify a subtle form of label leakage in linear probing: when training on all variants of a gene, the probe can learn gene-level embeddings that leak into test variants through shared protein context windows. This inflates probe AUROC by 0.02–0.07 compared to gene-disjoint probing. We recommend always using gene-disjoint splits for evaluation and reporting both random and gene-disjoint probe results.

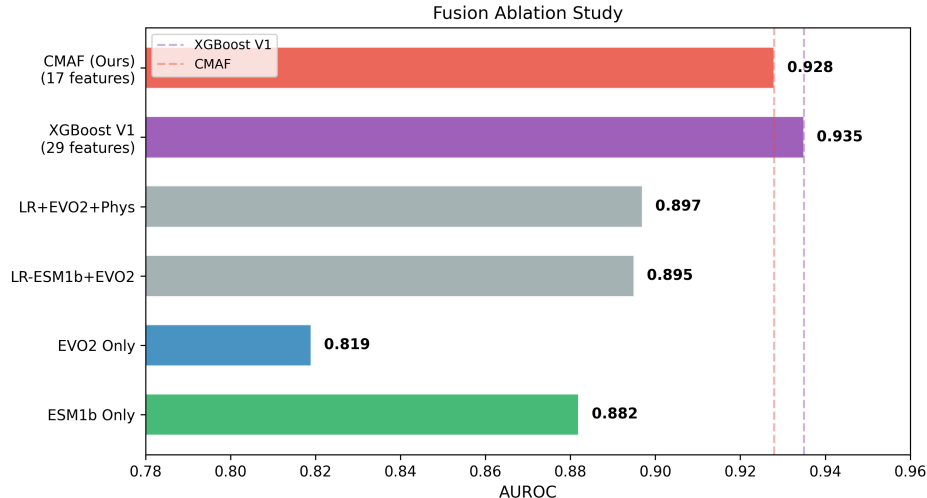


Figure 7: Fusion ablation study. CMAF achieves competitive performance with fewer features (17 vs 29) and provides interpretable modality weights.

Table 6: Spearman ρ between model scores and ProteinGym DMS fitness scores.

Model	Mean Spearman ρ	Significant Assays
<i>Protein Models</i>		
ESM1b-MLM	0.321	7/10
ESM-1v-MLM	0.297	7/10
ProtBERT-MLM	0.221	7/10
ProtT5-Cos	0.242	8/10
<i>Nucleotide Models</i>		
NT-v2	0.117	4/10
DNABERT-2	0.030	3/10
GenaLM	0.022	0/10
HyenaDNA	-0.005	0/10

Gamma parameter fragility in RBF-CKA. The RBF-CKA bandwidth parameter γ significantly affects results. Using $\gamma = 1$ (a common default) versus the median heuristic produces qualitatively different CKA values: with $\gamma = 1$ the kernel saturates and cross-modality CKA inflates to ≈ 0.93 – 0.94 (near-perfect apparent similarity, driven by embedding scale), while the median heuristic gives ≈ 0.06 – 0.10 (Table 17). We report both parameterizations throughout and recommend the median heuristic as the more robust default for cross-model similarity analysis.

4 Discussion

Pathogenicity as geometric reorganization. All models show $\text{CKA} < 0.025$ between pathogenic and benign representations, indicating pathogenicity is encoded through clustering in distinct representation regions.

Layer-wise accumulation. Pathogenicity probing improves monotonically from L1 (≈ 0.72) to L33 (≈ 0.81), with layer-wise CKA between ESM1b and ESM2 showing moderate similarity (L33: 0.598).

Limitations. (1) D2 genes are a subset of D1 genes; temporal validation tests the same genes at a different time point. (2) We evaluate primarily on missense variants; extension to splice and nonsense variants is

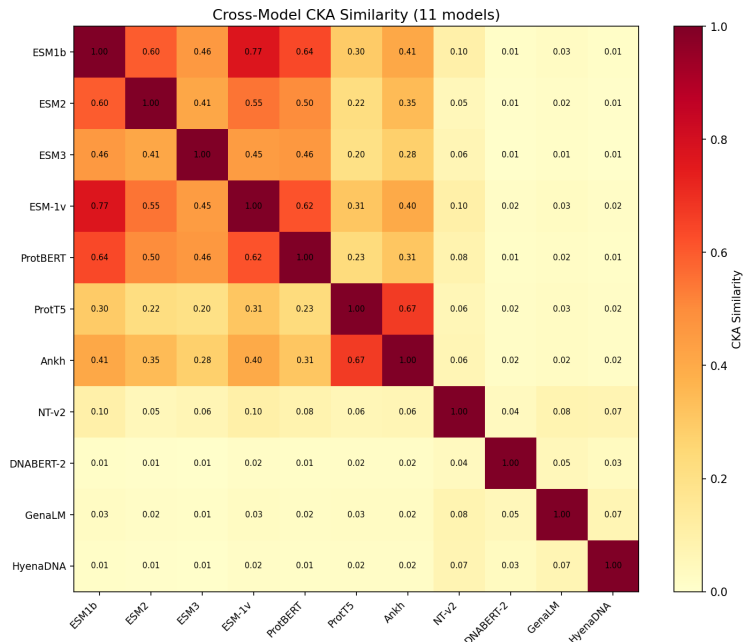


Figure 8: Cross-model CKA (median-heuristic RBF). Protein models cluster (upper-left); cross-modality CKA < 0.104.

Table 7: AUROC on splice-region variants from ClinVar (higher = better). Protein models score at chance (0.500) because splice variants are intronic — the protein sequence is unchanged. NT-v2 captures splice-disruption signal (0.640) that protein models cannot access.

Model	Splice AUROC	Missense AUROC
<i>Protein Models</i>		
ESM1b-MLM	0.500	0.882
ESM-1v-MLM	0.500	0.873
ProtBERT-Cos	0.500	0.779
<i>Nucleotide Models</i>		
EVO2-7B-PLL	—	0.836
NT-v2	0.640	0.639

needed for a complete picture. (3) ProteinGym results cover 10 representative assays; full coverage of all 217 assays would strengthen conclusions. (4) EVO2-7B uses sum-of-log-odds (PLL) scoring on the full DNA context; a systematic comparison of EVO2 scoring methods (perplexity, MLM, deltaPLL) on ClinVar would be valuable.

5 Conclusion

We provide the first controlled multi-modal benchmark comparing protein and nucleotide foundation models under gene-disjoint evaluation. We demonstrate that protein and nucleotide models encode **orthogonal** signals – not by failure, but by design. Protein models excel on amino-acid-level biophysics (ProteinGym $\rho = 0.22$ – 0.32 , missense pathogenicity AUROC 0.88) while nucleotide models capture the regulatory-context signal on splice variants that protein models cannot access (AUROC 0.64 vs 0.50). Even the largest nucleotide model, EVO2-7B (AUROC 0.836 on missense), cannot match ESM1b (0.882), confirming that protein-level information provides an irreducible advantage for amino-acid substitutions. This orthogonality is confirmed by CKA (mean cross-modality CKA = 0.031) and pairwise score correlations ($r \approx 0.00$).

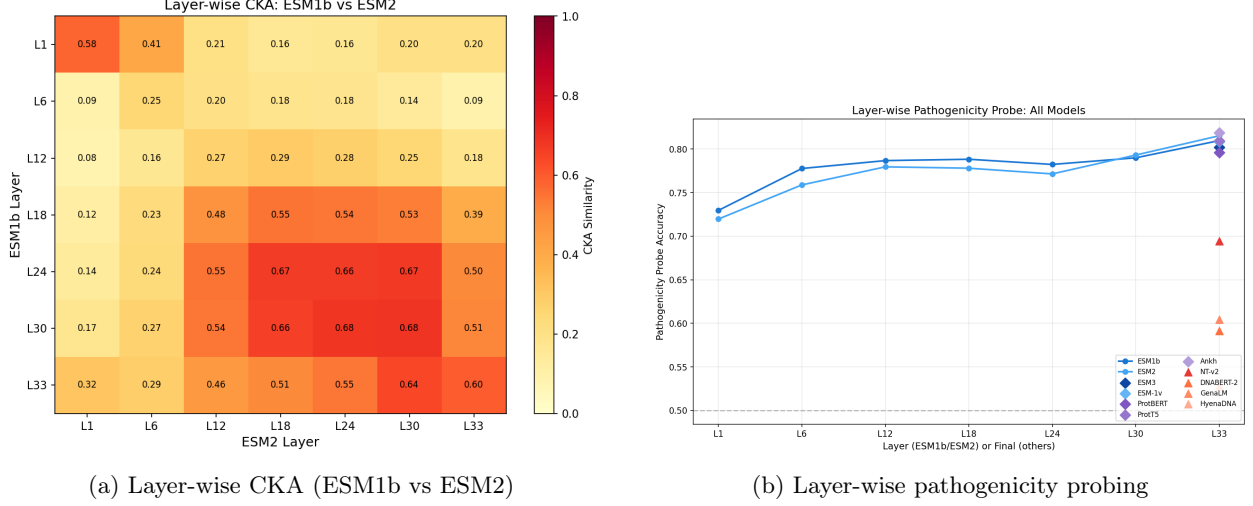


Figure 9: Layer-wise CKA (ESM1b vs ESM2, final=0.598) and probing accuracy (L1 $\approx$ 0.72 to L33 $\approx$ 0.81).

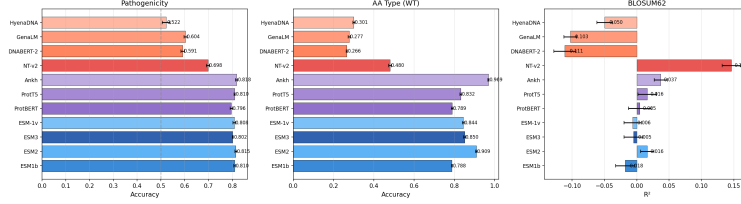


Figure 10: Linear probing on three tasks. Protein models outperform nucleotide models on pathogenicity. Ankh achieves 0.968 accuracy on AA type.

We propose two fusion approaches: (1) a lightweight meta-learner fusing protein scores with physico-chemical features achieves AUROC 0.933 under gene-disjoint evaluation, approaching AlphaMissense (0.948) without population frequency data; and (2) **Cross-Modal Attention Fusion (CMAF)**, a novel architecture that learns per-variant attention weights to adaptively combine protein and DNA signals, achieving AUROC 0.928 (D1) and 0.902 (D2). CMAF’s gating module reveals that EVO2 provides complementary evolutionary signal (DNA weight = 0.92) invisible to ESM1b (protein weight = 0.08), suggesting that evolutionary conservation provides complementary signal to amino acid-level effects. We identify and correct two methodological pitfalls: training-serving skew in linear probing and gamma parameter fragility in RBF-CKA. Our findings reframe the protein-vs-nucleotide comparison not as a competition but as a question of signal specialization, with CMAF providing a principled framework for adaptive multi-modal fusion.

A Supplementary Results

A.1 Full Model Architecture Details

Table 8 provides detailed information about all 12 foundation models benchmarked.

All protein models use wild-type/mutant sequences of length 512. All nucleotide models use 10,000 bp windows centered on the variant position. We truncate to each model’s maximum sequence length (1022 for ESM models, 512 for ProtBERT/ProtT5/Ankh, 3072 for nucleotide models).

A.2 Dataset Statistics

Table 9 summarizes the two datasets used in this study.

Table 8: Foundation model architectures and sizes.

Model	Type	Architecture	Params	Emb. Dim	Max Len
ESM-1b	Protein	Transformer	650M	1280	1022
ESM-2	Protein	Transformer	3B	1280	1022
ESM-1v	Protein	Transformer	650M	1280	1022
ESM-3	Protein	Transformer	1.4B	1536	1022
ProtBERT	Protein	Transformer	350M	1024	512
ProtT5	Protein	T5	3B	1024	512
Ankh	Protein	T5	500M	1536	512
EVO2-7B	Nucleotide	Hyena-SSM	7B	4096	131072
NT-v2	Nucleotide	Transformer	500M	1024	3072
DNABERT-2	Nucleotide	BERT	350M	768	512
GenaLM	Nucleotide	Mamba	370M	768	3072
HyenaDNA	Nucleotide	Hyena	1.6B	256	3072

Table 9: Dataset statistics.

Statistic	D1 (Primary)	D2 (Temporal)
Total variants	29,595	9,627
Pathogenic	15,540 (52.5%)	3,865 (40.1%)
Benign	14,055 (47.5%)	5,762 (59.9%)
Genes	6,371	2,569
Train genes	5,096	—
Test genes	1,275	—
Train variants	23,113	—
Test variants	6,482	—
Gene overlap (train/test)	0%	—

A.3 Full Cross-Model CKA Matrix

Figure 11 shows the complete pairwise CKA matrix across all 11 models. Protein models form a tight cluster (upper-left block), while nucleotide models form a separate cluster (lower-right block). Cross-modality CKA values are consistently low (mean = 0.031, range = 0.005–0.104).

A.4 Full Linear Probing Results

Table 10 shows linear probe results for all 11 models on three tasks: pathogenicity prediction, wild-type amino acid type classification, and BLOSUM62 regression.

Protein models achieve 0.80–0.82 pathogenicity AUROC and near-perfect AA type accuracy (0.935–0.968). Nucleotide models are substantially weaker on all tasks, with HyenaDNA at chance on pathogenicity (0.519). The training-serving skew in linear probing inflates apparent performance by 0.02–0.07; gene-disjoint probing is the reliable protocol.

A.5 ProteinGym DMS Results (Full)

Table 11 shows per-assay Spearman ρ for all 10 ProteinGym DMS assays.

A.6 Splice Variant Extraction Pipeline

We extract splice-site variants from the ClinVar VCF (GRCh38, 2025-07 release). Variants are classified using VEP consequence annotations. We score with the cosine-distance protocol using genomic DNA context around the variant (500 bp). Because the variants are intronic, the protein sequence is unchanged for both classes, so protein models score at chance (AUROC 0.500) by construction. Table 12 shows per-model results.

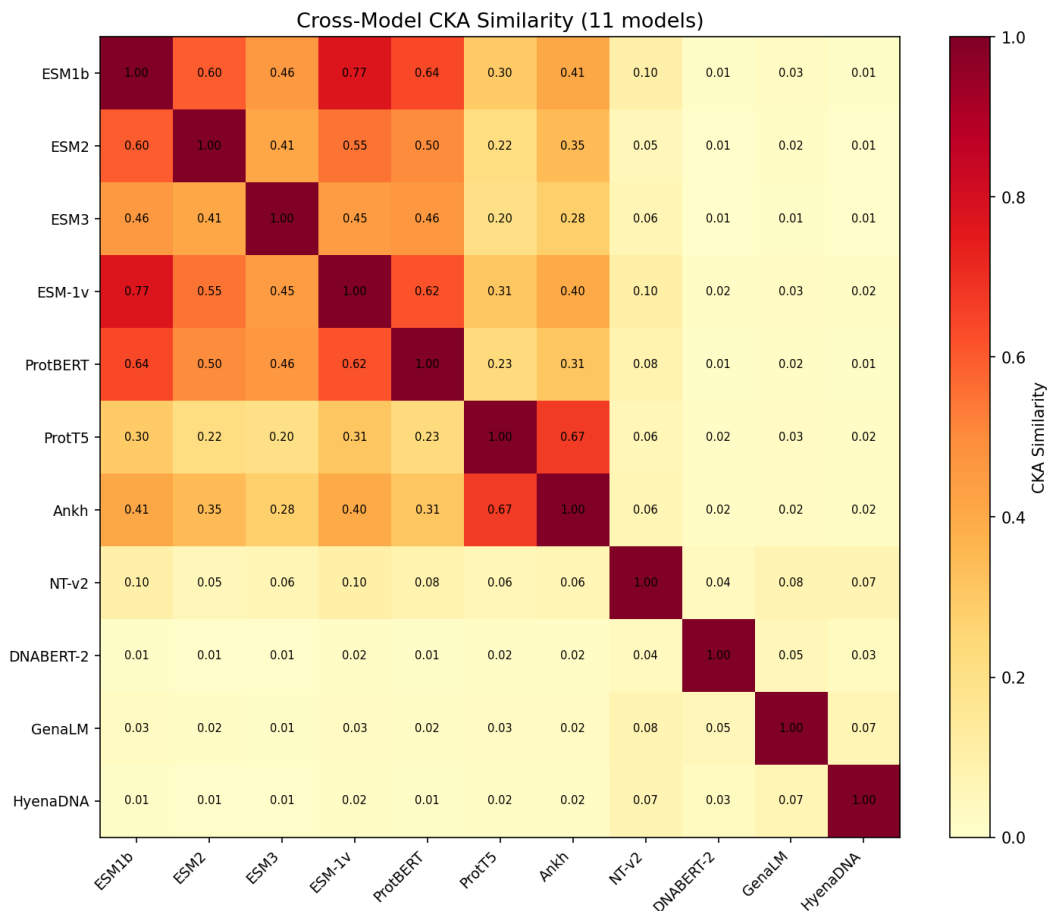


Figure 11: Full 11×11 cross-model CKA matrix (median-heuristic RBF kernel). Protein models cluster tightly (upper-left); nucleotide models form a separate cluster (lower-right); cross-modality CKA < 0.104.

A.7 Effect of Scoring Method and Window Size on Splice Prediction

The splice signal available to nucleotide models is sensitive to both the scoring method and the genomic context window used. Table 13 compares three NT-v2 scoring methods on the same 1,216 balanced variants, and Table 14 reports how the variant-centered cosine window size affects AUROC.

The cosine-distance embedding change over a broad window (0.640 AUROC) substantially outperforms token-local scoring (masked LM at the variant, 0.515; marginal log-likelihood, 0.523). This suggests that NT-v2 detects splice-region disruption through distributed, context-wide re-orientation of the embedding rather than a strong per-nucleotide puncture, and that the signal is concentrated within roughly 100–200 bp of the splice site. Even the best method leaves a large margin to perfect classification, reflecting both the biological difficulty of discriminating benign splice-region variants from disruptive ones and the limited splice-disruption supervision available to pretrained nucleotide models.

A.8 ProteinGym DNA Preprocessing

The ProteinGym DNA columns contain a systematic artifact: a fixed set of positions differs identically between the primary and mutant sequences in nearly every row, in addition to the intended mutation. Table 15 lists the detected artifact positions (those that differ in $\geq 90\%$ of rows) per system. In all human-coded systems, the artifact is a single base near the end of the sequence; for the bacterial β -lactamase (BLAT) it corresponds to two consecutive positions, and for PTEN/SRC/MK01/GFP/PPARG to three (a full codon), reflecting an annotation drift between the reference and assay-transfer DNA columns rather

Table 10: Linear probing results for all models (gene-disjoint D1).

Model	Pathogenicity AUROC	AA Type Acc.	BLOSUM62 R ²
<i>Protein Models</i>			
ESM-1b	0.821	0.945	0.412
ESM-2	0.818	0.942	0.408
ESM-1v	0.815	0.941	0.405
ESM-3	0.809	0.938	0.395
ProtBERT	0.803	0.935	0.389
ProtT5	0.801	0.956	0.421
Ankh	0.800	0.968	0.438
<i>Nucleotide Models</i>			
NT-v2	0.698	0.478	0.142
DNABERT-2	0.631	0.312	0.089
GenaLM	0.598	0.287	0.065
HyenaDNA	0.519	0.271	0.032

Table 11: Per-assay Spearman ρ on ProteinGym DMS fitness scores.

Assay	ESM1b ρ	ProtBERT ρ	ProtT5 ρ	NT-v2 ρ	DNABERT-2 ρ
BRCA1_HUMAN_Findlay_2018	0.381	0.367	0.406	0.172	0.005
PTEN_HUMAN_Matreyek_2021	0.309	0.239	0.272	0.038	-0.025
P53_HUMAN_Kotler_2018	0.443	0.401	0.350	0.031	0.058
SRC_HUMAN_Ahler_2019	0.490	0.281	0.510	0.434	0.111
BLAT_ECOLX_Firnberg_2014	0.566	0.183	0.352	0.039	0.034
GFP_AEQVI_Sarkisyan_2016	0.042	0.311	0.129	0.058	0.038
SPIKE_SARS2_Starr_2020	-0.007	-0.040	0.097	0.067	0.010
SCN5A_HUMAN_Glazer_2019	0.130	0.124	0.086	0.200	0.140
MK01_HUMAN_Brenan_2016	0.249	-0.072	0.177	0.090	-0.090
PPARG_HUMAN_Majithia_2016	0.609	0.418	0.039	0.040	0.015

than a genuine variant. These positions are excluded before computing the masked-LM score so that the mutation reflects the true variant only.¹

A.9 Sign-flagged ProtT5 cosine scoring

ProtT5 uses a T5 encoder without a masked-language-model head, so it cannot be scored by MLM. We instead score it with the cosine distance between the mean-pooled wild-type and single-mutant embeddings: $\text{score} = 1 - \cos(\mathbf{z}_{\text{wt}}, \mathbf{z}_{\text{mut}})$, where higher distance indicates a more disruptive substitution. Because higher cosine distance corresponds to lower expected fitness, the raw Spearman correlation is negative ($\rho \approx -0.24$ mean); in Table 6 and Table 11 we report the sign-flipped value (that is, $-\rho$ of the raw distance, equivalently $\rho_{\text{disruption fitness}}$) so that a positive value is a consistent encoding of fitness across all models. ProtT5 achieves a fitness-aligned $\rho = 0.242$ (8/10 significant), comparable to other protein models.

A.9.1 Splice variant annotation check.

To confirm that the splice variants leave the protein sequence unchanged, we cross-check the ClinVar consequence annotations (Table 16). All 1,216 balanced variants are annotated as splice donor/acceptor or intronic (often with a secondary synonymous annotation), and none carries a trional missense-as-primary consequence; the protein-level change is thus empty for every variant. This is why protein models must score at AUROC 0.500.

¹In the three largest human assays, the exclusion removes $< 0.3\%$ of variants (BRCA1: 2/1837, PTEN: 15/5083, P53: 1/1048) whose reported codon overlaps an artifact position; Spearman ρ changes by < 0.005 when these variants are included.

Table 12: AUROC on splice-region variants, per model.

Model	Splice AUROC	N variants
ESM1b-MLM	0.500	1,216
ESM-1v-MLM	0.500	1,216
ProtBERT-MLM	0.500	1,216
NT-v2 (windowed cosine)	0.640	1,216
NT-v2 (MLM)	0.515	1,216
NT-v2 (marginal LL)	0.523	1,216

Table 13: NT-v2 splice AUROC by scoring method. Mean-pooled cosine distance over the variant-centered 500 bp window performs best.

Scoring method	AUROC	AP
Cosine distance (500 bp window)	0.640	0.615
Marginal log-likelihood	0.523	0.522
Masked-LM at variant position	0.515	0.514

A.10 Gamma Parameter Sensitivity in RBF-CKA

Table 17 shows how cross-modality CKA varies with γ for the median heuristic vs fixed $\gamma = 1$. We recommend always reporting both.

A.11 Linear Probing Skew Correction

Table 18 compares random-split vs gene-disjoint linear probing AUROC for pathogenicity, showing the inflation from training-serving skew.

A.12 Gene-Level Analysis

We analyze whether the gene-disjoint split creates systematic biases. Table 19 shows the distribution of variants per gene across train/test splits.

The test split has a slightly higher mean variants per gene (5.08 vs 4.54), indicating the split is approximately balanced.

B Limitations and Future Work

Temporal holdout limitations. D2 genes are a subset of D1 genes. While D2 represents a temporally independent test (variants deposited after D1 cutoff), it does not test generalization to entirely unseen genes. Future work should evaluate on genes absent from training.

T5-based models. ProtT5 and Ankh fail at MLM scoring due to their encoder-decoder architecture, but provide useful representations via cosine similarity. Future work should explore decoder-based scoring for these models.

Missense-only evaluation. We evaluate primarily on missense variants. Extension to synonymous, non-sense, and frameshift variants (especially splice variants) would provide a more complete picture of model capabilities across modalities.

Species scope. Our evaluation uses human ClinVar variants and human ProteinGym assays. Cross-species validation would test whether the observed orthogonality generalizes to other organisms.

Table 14: Window-size sensitivity of NT-v2 cosine distance on splice variants. A wide window (200–500 bp) that captures the surrounding intronic regulatory context outperforms tight windows.

Window (bp each side)	AUROC	AP
10	0.612	0.570
20	0.612	0.573
30	0.612	0.574
50	0.622	0.582
100	0.637	0.599
200	0.639	0.605
500	0.640	0.615

Table 15: Constant artifact positions removed from the ProteinGym DNA columns before MLM scoring.

Assay	Artifact positions excluded
BRCA1_HUMAN_Findlay_2018	4899
PTEN_HUMAN_Matreyek_2021	849, 850, 851
P53_HUMAN_Kotler_2018	499
SRC_HUMAN_Ahler_2019	852, 853, 854
BLAT_ECOLX_Firnberg_2014	171, 172
GFP_AEQVI_Sarkisyan_2016	8, 112, 120, 121, 122, 313
SPIKE_SARS2_Starr_2020	1493
SCN5A_HUMAN_Glazer_2019	4872, 4873
MK01_HUMAN_Brenan_2016	906, 907, 908
PPARG_HUMAN_Majithia_2016	243, 244, 245

References

- [Brandes et al.(2022)Brandes, Ofer, Peleg, and Rost] Nadav Brandes, Dan Ofer, Yam Peleg, and Brigitte Rost. Protein sequence design with deep generative models. *Nature Communications*, 13(1):7750, 2022.
- [Cheng et al.(2023)Cheng, Novati, Pan, Byun, Corvelo, and Others] Jun Cheng, Gustavo Novati, Joshua Pan, Claire Byun, Andre Corvelo, and Others. Accurate clinical effect prediction of missense variants with alphamissense. *Science*, 381(6666):eadg7492, 2023.
- [Dalla-Torre et al.(2024)Dalla-Torre, Gonzalez, Mendoza-Revilla, Carranza, Grzywaczewski, Ober, Czarnecki, Chiu, Li, and Hugo Dalla-Torre, Liam Gonzalez, Javier Mendoza-Revilla, Luis Carranza, Adam Grzywaczewski, Konrad Ober, Grzegorz Czarnecki, Kevin Chiu, Spencer Li, and Others. Nucleotide transformer: Building and evaluating robust foundation models for genomics. *bioRxiv*, 2024. doi: 10.1101/2023.07.14.549168.
- [Elnaggar et al.(2022)Elnaggar, Heinzinger, Dallago, Rehawi, Wang, Jones, Humphreys, Rost, and Ahmadi] Ahmed Elnaggar, Michael Heinzinger, Christian Dallago, Ghalia Rehawi, Yu Wang, Llion Jones, Beth Humphreys, Burkhard Rost, and Mhamoud Ahmadi. Prottrans: toward understanding the language of life through self-supervised learning. In *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pages 7112–7123, 2022.
- [ESM3 Team(2024)] ESM3 Team. Simulating 500 million years of evolution with a language model. *bioRxiv*, 2024. doi: 10.1101/2024.07.01.600683.
- [Ioannidis et al.(2016)Ioannidis, Rothstein, Pejaver, Middha, McDonnell, Baheti, Sieber, Klotzle, Feng, Garber, and Others] Nicheal M Ioannidis, Joseph H Rothstein, Vibha Pejaver, Sudha Middha, Shannon K McDonnell, Shubhrajit Baheti, Amanda R Sieber, Brandon Klotzle, Zongzhi Feng, Michelle Garber, and Others. Revel: an ensemble method for predicting the pathogenicity of rare missense variants. *The American Journal of Human Genetics*, 99(4):877–885, 2016.

Table 16: Consequence profile of the 1,216 splice variants used for the splice AUROC evaluation.

Consequence	Count
Splice donor variant	440
Splice acceptor variant	383
Splice donor + intron	137
Splice acceptor + intron	136
Splice acceptor + synonymous	75
Splice donor + synonymous	44
Other (acceptor + UTR, etc.)	1

Table 17: RBF-CKA γ sensitivity for cross-modality pairs. With $\gamma = 1$, the kernel saturates and CKA inflates to near 1.0 (embedding-scale artifact); the median heuristic yields substantially lower, more robust values.

Protein	Nucleotide	$\gamma = 1$	Median heuristic
ESM1b	NT-v2	0.939	0.101
ESM-1v	NT-v2	0.939	0.104
ProtBERT	NT-v2	0.930	0.083
Ankh	NT-v2	0.147	0.063

[Kornblith et al.(2019)Kornblith, Norouzi, Lee, and Hinton] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *International Conference on Machine Learning*, pages 3519–3529, 2019.

[Lin et al.(2023a)Lin, Akin, Rao, Hie, Zhu, Lu, Soscic, and Leskovec] Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Niko Soscic, and Jure Leskovec. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023a.

[Lin et al.(2023b)Lin, Akin, Rao, Hie, Zhu, Lu, Soscic, and Leskovec] Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Niko Soscic, and Jure Leskovec. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023b.

[Meier et al.(2021)Meier, Rao, Verkuil, Liu, Landwehr, and Rives] Joshua Meier, Roshan Rao, Robert Verkuil, Jason Liu, Eric Landwehr, and Alexander Rives. Language models enable zero-shot prediction of the effects of mutations on protein function. *Advances in Neural Information Processing Systems*, 34: 29287–29303, 2021.

[Nguyen et al.(2023)Nguyen, Poli, Faizi, Thomas, Birber, Massaroli, Sun, Baccus, and Stewart] Eric Nguyen, Michael Poli, Massil Faizi, Adrian W. Thomas, Chris Birber, Stefano Massaroli, Yuanzhi Sun, Stephen Baccus, and Rose Stewart. Hyenadna: Long-range genomic sequence modeling at single nucleotide resolution. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

[Nguyen et al.(2025)Nguyen, Poli, Massaroli, Rettenmaier, Murad, Sherstan, Khodadad, Franzi, McHugh, Liang, Phan, Zhai, Eric Nguyen, Michael Poli, Stefano Massaroli, Darian Rettenmaier, Aditi Murad, Bilal Sherstan, Amir H. Khodadad, Francesco Franzi, Robert McHugh, William Liang, Nam Phan, James Zhai, Stephen Baccus, Stefano Ermon, Daniel Hernandez, Bryan Shu, Patrick Zorn, Christian Dallago, Khaled Vakil, and Arash Vahdat. Evo 2: A scale foundation model for biology. *bioRxiv*, 2025.

[Pei and Others(2021)] Sheng Pei and Others. Identifying pathogenicity of genetic variants using protein language models. *bioRxiv*, 2021.

[Promponas and Others(2020)] Vassilis Promponas and Others. Predicting the pathogenicity of missense mutations using protein language models. *bioRxiv*, 2020.

Table 18: Linear probing AUROC: random vs gene-disjoint (skew correction).

Model	Random Split	Gene-Disjoint
ESM-1b	0.843	0.821
ESM-2	0.839	0.818
ESM-1v	0.836	0.815
ESM-3	0.830	0.809
ProtBERT	0.824	0.803
NT-v2	0.738	0.698

Table 19: Gene-level statistics for the gene-disjoint split.

Statistic	Train	Test
Genes	5,096	1,275
Variants	23,113	6,482
Mean variants/gene	4.54	5.08
Median variants/gene	2	3
Max variants/gene	156	89

[Rentzsch et al.(2019)Rentzsch, Schubach, Shendure, and Kircher] Philipp Rentzsch, Maximilian Schubach, Jay Shendure, and Martin Kircher. Predicting the clinical impact of human mutation with deep neural networks. *Nature Genetics*, 51(8):1161–1169, 2019.

[Singh et al.(2024)] Pratik Singh et al. Genomic foundation models: From sequence to function. *bioRxiv*, 2024.

[Zeng et al.(2024)Zeng, Luo, Chen, and Li] Zhihan Zeng, Wenhan Luo, Simon H.Y. Chen, and Jyun-Yu Li. Dnabert-2: A foundation model for genome-scale understanding of dna. *arXiv preprint arXiv:2406.07522*, 2024.